MARCEL BASUMATARY

Data Scientist

Phone: +91-7086964899 **Email:** marcelbasumatary5@gmail.com **LinkedIn:** linkedin.com/in/marcel-basumatary

GitHub: github.com/Marcel-header **Nationality:** Indian

Location: Assam, India (Requires Visa Sponsorship / Austrian RWR Card for Vienna)

Data Scientist (B.Tech, 2026) specializing in sensor data analytics, anomaly detection, and end-to-end data pipelines for automated systems. Co-author of an IEEE-published framework where I processed 10,600+ records to detect complex behavioral anomalies using spatial AI. Experienced in building structured data collection pipelines, analyzing time-series and geospatial IoT data (GPS geofencing), and developing system performance dashboards using Python (NumPy, Pandas) and SQL. Passionate about applying predictive analytics and signal processing concepts to optimize human-centric automation at Iono.

CORE TECHNICAL COMPETENCIES

Data & Sensor Analytics: Python (NumPy, Pandas, FastAPI), SQL (PostgreSQL), Time-Series Analysis, IoT/GPS Sensor Analytics, Signal Processing Concepts, System Log Analysis.

ML & Predictive Modeling: Anomaly Detection, Predictive Maintenance Concepts, Scikit-learn, PyTorch (facenet-pytorch), Pose Estimation, Model Outcome Validation.

Pipelines & Annotation: Structured Data Collection, Data Annotation Workflows, Data Cleaning, Automated Preprocessing, Large-scale Datasets.

Dashboards & Production: Data Visualization (Matplotlib, React.js), REST APIs (FastAPI, Swagger UI), Docker, Git, Cross-Architecture Benchmarking.

ML RESEARCH & DATA ENGINEERING INITIATIVES

Lead ML Researcher & Co-Author — Anomaly Detection Pipeline | ***Kaziranga University (IEEE ICC-CNS 2026)*** **Aug 2025 – May 2026**

- **Anomaly Detection & Sensor Processing:** Architected a 4-stage sequential ML pipeline to detect behavioral anomalies (traffic violations) from continuous video feeds. Engineered structured data pipelines to process visual sensor data and compressed the model to ~6 MB for real-time edge evaluation.
- **Data Annotation & Processing:** Curated, cleaned, and structured a large-scale dataset of 10,600+ records. Managed complex data annotation workflows and preprocessing using Python (NumPy, Pandas) to ensure robust ground-truth labeling and model training.
- **System Performance & Validation:** Analyzed system performance data and validated model outcomes, achieving 99.5% mAP@0.5. Conducted rigorous statistical evaluations of precision, recall, and stability across continuous frame sequences (time-series evaluation).
- **Predictive Dashboards:** Built data visualization dashboards (Matplotlib) to track performance metrics and deployed the predictive inference engine via a production Flask REST API.

PROFESSIONAL EXPERIENCE

Data Engineering & AI Intern (IoT & System Analytics) | ***NIELIT (National Institute of Electronics & IT), Tezpur*** **Aug 2026 – Present**

- **IoT Sensor Analytics & Algorithms:** Engineered a 3-layer anomaly verification pipeline integrating spatial IoT data (100m GPS geofencing via the Haversine formula) and time-window logs to enforce location-verified outcomes.
- **Biometric Data & APIs:** Developed high-performance backend pipelines using FastAPI, deploying an offline facial recognition model (facenet-pytorch, MTCNN, InceptionResnetV1) to process and validate 512-Dimensional embeddings using cosine similarity.
- **Structured Data Collection & Dashboards:** Built full-stack administrative dashboards (React.js, Python) to centralize real-time sensor streams into Neon PostgreSQL, continuously validating edge-device accuracy for non-technical stakeholders.

PEER-REVIEWED PUBLICATIONS, EDUCATION & CREDENTIALS

Published Co-Author — IEEE Co-Sponsored ICC-CNS 2026 | ***Cert No: VFSTR/CSE/CNS1521/5*** **June 2026**

S. Gangmei, D. Das, B. Paul, S. Hazarika, **M. Basumatary**, V. Chusi, C. Borah. *"Modular AI-Based Framework for Two-Wheeler Traffic Violation Detection Using Object Detection, Pose Estimation, and ALPR."* Open-Source Repository: `github.com/Marcel-header/traffic-violation-detection-yolov5-AI-Traffic-Enforcement-System`